j4_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)

```
         LEFT rail                            RIGHT rail
                             +--[ USB-C ]--+
                  3.3V -- 3V3|             |  5V
                ground -- GND|             |  GND  ground
                ground -- GND|     TTGO    |  27   DISP-L RX <- XIAO D6
   (avoid: boot/flash) -- 12 |  T-Display  |  26   DISP-R RX <- XIAO D6
        EYE POP toggle -- 13 |     v1.1    |  25   DISP-R TX (face msgs)
            AUX toggle -- 15 |             |  33   VENT toggle
                (free) -- 2  |   [ ST7789  |  32   LASER toggle
  DISP-L TX -> XIAO D7 -- 17 |    TFT on   |  39   (input only)
               I2C SCL -- 22 |    back ]   |  38   (input only)
               I2C SDA -- 21 |             |  37   (input only)
                ground -- GND|             |  36   (input only)
                ground -- GND|             |  3.3V
                             +-------------+

on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO0,
    GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
  I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, ADS_03 @ 0x4A, ADS_04 @ 0x4B,
    PCF8574 keypad_left @ 0x21, keypad_right @ 0x20
  toggles: INPUT_PULLUP, switch closes to GND (ON = LOW)
```

GPIO	Function
21	I2C SDA (ADS1115 x4, PCF8574 keypads x2)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, face-preset messages)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
32	LASER toggle (INPUT_PULLUP, switch closes to GND)
33	VENT toggle (INPUT_PULLUP, switch closes to GND)
13	EYE POP toggle (INPUT_PULLUP, switch closes to GND)
15	AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in; data / MAC / connection status)

j4_controller -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

```
ADS_01 @ 0x48 (ADDR -> GND)                 both joysticks
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  GND  = address 0x48
| ALRT |  not connected
| A0   |  neck joystick X wiper        -> neck L/R differential mix
| A1   |  neck joystick Y (jaw) wiper  -> neck L/R base height
| A2   |  eyes joystick X wiper        -> eyes pan servo  (PCA9685 ch 3)
| A3   |  eyes joystick Y wiper        -> eyes tilt servo (PCA9685 ch 4)
+------+

ADS_02 @ 0x49 (ADDR -> VDD)                 four face pots, left bank
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  VDD  = address 0x49
| ALRT |  not connected
| A0   |  Eyebrow L pot wiper          -> PCA9685 ch 6
| A1   |  Eyebrow R pot wiper          -> PCA9685 ch 7
| A2   |  Basket Eyebrow L pot wiper   -> PCA9685 ch 8
| A3   |  Basket Eyebrow R pot wiper   -> PCA9685 ch 9
+------+

ADS_03 @ 0x4A (ADDR -> SDA)                 three face pots, right bank
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  SDA  = address 0x4A
| ALRT |  not connected
| A0   |  Nose pot wiper (up/down)     -> PCA9685 ch 10
| A1   |  FAULTY channel, nothing connected, left unread in firmware
| A2   |  Bottom Eyelid L pot wiper    -> PCA9685 ch 12
| A3   |  Bottom Eyelid R pot wiper    -> PCA9685 ch 13
+------+

ADS_04 @ 0x4B (ADDR -> SCL)                 neck-pivot + faders + Nose Basket
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  SCL  = address 0x4B
| ALRT |  not connected
| A0   |  neck-pivot pot wiper         -> nP on the stepper link (0-3200)
| A1   |  fader_left wiper             -> neck pivot (doubles ADS_04 A0, 0-3200)
| A2   |  fader_right wiper            -> iris (doubles j4_display_right IRIS)
| A3   |  Nose Basket pot wiper        -> PCA9685 ch 11 (single source)
+------+
```